

11 (**short**) A micromachined accelerometer embedded in a smartphone is represented by a mass-spring-dashpot system as shown schematically in Fig. 8. The natural frequency for this system is 1000 Hz.

(a) The maximum overshoot in the accelerometer response when the smartphone is subject to unit step input in acceleration is observed to be 25% higher than the final nominal value. Estimate the damping factor and logarithmic decrement of the device. [5]

(b) The phone is left on a machine vibrating at 1000 Hz subjecting the device to a base acceleration,  $a$ , of amplitude  $1 \text{ m s}^{-2}$  at this frequency. Estimate the magnitude of the steady-state displacement of the device. [5]

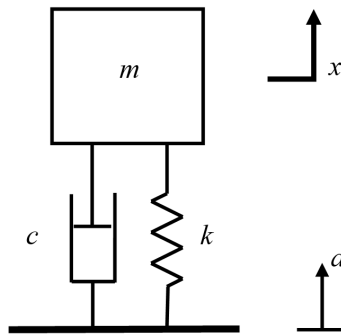


Fig. 8